

GED Math Answer Guide: Number Sense and Operations

Detailed Explanations

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1 Basic Level (Questions 1-10)

Question 1: Order of Operations

Correct Answer: 21

Problem

Calculate: $8 + 3 \times 5 - 2 = ?$

Step-by-Step Solution

1. Apply the order of operations (PEMDAS/BODMAS)
2. Multiplication first: $3 \times 5 = 15$
3. Rewrite the expression: $8 + 15 - 2$
4. Addition and subtraction (left to right): $8 + 15 = 23$
5. Final subtraction: $23 - 2 = 21$

Mathematical Concepts

Order of operations ensures consistent calculation results. PEMDAS stands for Parentheses, Exponents, Multiplication and Division (left to right), Addition and Subtraction (left to right).

Common Mistakes

- Adding $8 + 3$ first, then multiplying by 5
- Subtracting before adding
- Ignoring the order of operations entirely

How to Avoid

Memorize PEMDAS and always check for multiplication/division before addition/subtraction.

Alternative Method/Verification

Use a calculator to verify: $8 + 3 \times 5 - 2 = 21$

Real-World Context

Budgeting: If you have \$8, earn \$3 per hour for 5 hours, then spend \$2, you have \$21 total.

Question 2: Adding Fractions

Correct Answer: $\frac{7}{12}$

Problem

Calculate: $\frac{1}{4} + \frac{1}{3} = ?$

Step-by-Step Solution

1. Find the least common denominator (LCD) of 4 and 3
2. $\text{LCD} = 12$
3. Convert $\frac{1}{4}$ to twelfths: $\frac{1}{4} = \frac{3}{12}$
4. Convert $\frac{1}{3}$ to twelfths: $\frac{1}{3} = \frac{4}{12}$
5. Add the fractions: $\frac{3}{12} + \frac{4}{12} = \frac{7}{12}$

Mathematical Concepts

When adding fractions, denominators must be the same. The LCD is the smallest number divisible by both denominators.

Common Mistakes

- Adding numerators and denominators separately: $\frac{1+1}{4+3} = \frac{2}{7}$ (incorrect)
- Using 7 as the common denominator instead of 12

How to Avoid

Always find the LCD first. Verify that both fractions have the same denominator before adding.

Alternative Method/Verification

Convert to decimals: $0.25 + 0.333... \approx 0.583$, and $\frac{7}{12} \approx 0.583$

Real-World Context

Cooking: Using $\frac{1}{4}$ cup of flour and $\frac{1}{3}$ cup of sugar gives $\frac{7}{12}$ cup total ingredients.

Question 3: Subtracting Decimals

Correct Answer: 3.47

Problem

Calculate: $8.92 - 5.45 = ?$

Step-by-Step Solution

1. Align decimal points:

$$\begin{array}{r} 8.92 \\ -5.45 \\ \hline \end{array}$$

2. Subtract from right to left: $2 - 5$ requires borrowing
3. Borrow 1 from the tenths place: $12 - 5 = 7$
4. Tenths place: $8 - 4 = 4$
5. Ones place: $8 - 5 = 3$
6. Result: 3.47

Mathematical Concepts

Decimal subtraction follows the same rules as whole number subtraction. Alignment of decimal points is critical for correct computation.

Common Mistakes

- Misaligning decimal points
- Forgetting to borrow when needed
- Subtracting the larger digit from the smaller

How to Avoid

Always line up decimal points vertically. Use borrowing notation clearly.

Alternative Method/Verification

Check by addition: $3.47 + 5.45 = 8.92$

Real-World Context

Money: If you spend \$5.45 from \$8.92, you have \$3.47 remaining.

Question 4: Converting Fractions to Decimals

Correct Answer: 0.375

Problem

Convert $\frac{3}{8}$ to a decimal.

Step-by-Step Solution

1. Divide the numerator by the denominator: $3 \div 8$
2. Long division:
$$3.000 \div 8 = 0.375$$
3. 8 goes into 30 three times ($8 \times 3 = 24$), remainder 6
4. Bring down 0: $60 \div 8 = 7$ ($8 \times 7 = 56$), remainder 4
5. Bring down 0: $40 \div 8 = 5$ ($8 \times 5 = 40$), remainder 0
6. Result: 0.375

Mathematical Concepts

Any fraction can be converted to a decimal by dividing the numerator by the denominator. Some decimals terminate (like 0.375), while others repeat.

Common Mistakes

- Dividing the denominator by the numerator (incorrect direction)
- Stopping division too early
- Misplacing the decimal point

How to Avoid

Remember: fraction bar means division. Numerator goes inside the division bracket, denominator outside.

Alternative Method/Verification

Multiply both numerator and denominator to get a power of 10 in the denominator:

$$\frac{3}{8} = \frac{3 \times 125}{8 \times 125} = \frac{375}{1000} = 0.375$$

Real-World Context

Measurements: $\frac{3}{8}$ inch equals 0.375 inches, useful for converting between fractional and decimal measurements.

Question 5: Finding Percentages

Correct Answer: 25%

Problem

15 is what percent of 60?

Step-by-Step Solution

1. Set up the percentage formula:

$$\text{Percentage} = \frac{\text{Part}}{\text{Whole}} \times 100\%$$

2. Substitute values:

$$\text{Percentage} = \frac{15}{60} \times 100\%$$

3. Simplify the fraction: $\frac{15}{60} = \frac{1}{4}$

4. Calculate: $\frac{1}{4} \times 100\% = 25\%$

Mathematical Concepts

A percentage represents a portion out of 100. The formula is: $\frac{\text{Part}}{\text{Whole}} \times 100\% = \text{Percentage}$

Common Mistakes

- Dividing 60 by 15 instead of 15 by 60
- Forgetting to multiply by 100
- Misidentifying the part and the whole

How to Avoid

Remember: the part (smaller number) goes on top, the whole (larger number) on the bottom.

Alternative Method/Verification

Use proportion: $\frac{15}{60} = \frac{x}{100}$, so $x = 25$

Real-World Context

Sales: 15 out of 60 items sold represents a 25% sales conversion rate.

Question 6: Multiplying Fractions

Correct Answer: $\frac{2}{15}$

Problem

Calculate: $\frac{2}{3} \times \frac{1}{5} = ?$

Step-by-Step Solution

1. Multiply numerators: $2 \times 1 = 2$
2. Multiply denominators: $3 \times 5 = 15$
3. Result: $\frac{2}{15}$
4. Check if simplifiable: $\text{GCD}(2, 15) = 1$, so $\frac{2}{15}$ is in simplest form

Mathematical Concepts

To multiply fractions, multiply numerators together and denominators together. Always simplify to lowest terms.

Common Mistakes

- Adding instead of multiplying
- Finding a common denominator (only needed for addition/subtraction)
- Not simplifying the final answer

How to Avoid

Use the rule: "multiply straight across." Check if common factors exist before multiplying to simplify early.

Alternative Method/Verification

Convert to decimals: $0.667 \times 0.2 = 0.133\dots$, and $\frac{2}{15} = 0.133\dots$

Real-World Context

Recipes: $\frac{2}{3}$ of $\frac{1}{5}$ cup of salt equals $\frac{2}{15}$ cup.

Question 7: Dividing Fractions

Correct Answer: $\frac{8}{9}$

Problem

Calculate: $\frac{2}{3} \div \frac{3}{4} = ?$

Step-by-Step Solution

1. Write the division problem: $\frac{2}{3} \div \frac{3}{4}$
2. Flip (invert) the second fraction: $\frac{3}{4}$ becomes $\frac{4}{3}$
3. Change division to multiplication: $\frac{2}{3} \times \frac{4}{3}$
4. Multiply numerators: $2 \times 4 = 8$
5. Multiply denominators: $3 \times 3 = 9$
6. Result: $\frac{8}{9}$

Mathematical Concepts

Dividing by a fraction is the same as multiplying by its reciprocal. The reciprocal of $\frac{a}{b}$ is $\frac{b}{a}$.

Common Mistakes

- Flipping the first fraction instead of the second
- Forgetting to invert the divisor entirely
- Adding or subtracting instead of multiplying after inversion

How to Avoid

Use the phrase: “Keep, Change, Flip”—keep the first fraction, change $\div$ to $\times$, flip the second fraction.

Alternative Method/Verification

Multiply by reciprocal: $\frac{2}{3} \times \frac{4}{3} = \frac{8}{9}$

Real-World Context

Sharing: If you have $\frac{2}{3}$ of a pizza and divide it into portions of $\frac{3}{4}$, you get $\frac{8}{9}$ portions.

Question 8: Exponents

Correct Answer: 32

Problem

Calculate: $2^5 = ?$

Step-by-Step Solution

1. Understand exponent notation: 2^5 means 2 multiplied by itself 5 times
2. Write out the multiplication: $2 \times 2 \times 2 \times 2 \times 2$
3. Calculate step by step:

$$2 \times 2 = 4$$

$$4 \times 2 = 8$$

$$8 \times 2 = 16$$

$$16 \times 2 = 32$$

4. Result: 32

Mathematical Concepts

An exponent (or power) tells how many times the base is multiplied by itself. In 2^5 , the base is 2 and the exponent is 5.

Common Mistakes

- Multiplying the base by the exponent: $2 \times 5 = 10$ (incorrect)
- Computing only partial repetitions
- Confusing exponents with coefficients

How to Avoid

Expand the expression fully and multiply step by step. Use a calculator for verification.

Alternative Method/Verification

Use a calculator: $2^5 = 32$

Real-World Context

Data storage: A computer address with 2^5 possible values has 32 unique combinations.

Question 9: Square Roots

Correct Answer: 8

Problem

Calculate: $\sqrt{64} = ?$

Step-by-Step Solution

1. Understand square root: Find the number that, when multiplied by itself, equals 64
2. Think: What times what equals 64?
3. Test: $8 \times 8 = 64$
4. Result: $\sqrt{64} = 8$

Mathematical Concepts

A square root is the inverse operation of squaring. If $a^2 = b$, then $\sqrt{b} = a$.

Common Mistakes

- Dividing by 2: $64 \div 2 = 32$ (incorrect)
- Confusing square root with division by 2
- Getting negative square roots (though -8 is also technically valid)

How to Avoid

Learn perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, ...

Alternative Method/Verification

Check by squaring: $8^2 = 8 \times 8 = 64$

Real-World Context

Geometry: A square with area 64 square feet has sides of length 8 feet.

Question 10: Absolute Value

Correct Answer: 5

Problem

Calculate: $|-5| = ?$

Step-by-Step Solution

1. Understand absolute value: It is the distance from zero on a number line, always positive
2. The number -5 is 5 units away from zero
3. Result: $|-5| = 5$

Mathematical Concepts

Absolute value $|x|$ is always non-negative. It represents the magnitude or distance from zero, regardless of direction.

Common Mistakes

- Returning the negative value: $|-5| = -5$ (incorrect)
- Confusing absolute value with the number itself

How to Avoid

Remember: absolute value makes all numbers positive (or zero).

Alternative Method/Verification

Check: $|-5| = |(-1) \times 5| = 1 \times |5| = 5$

Real-World Context

Temperature: An absolute value of -5 represents 5 degrees below zero.

2 Intermediate Level (Questions 11-20)

Question 11: Complex Order of Operations

Correct Answer: 2

Problem

Calculate: $24 \div (4 + 2) - 3 + 1 = ?$

Step-by-Step Solution

1. Parentheses first: $4 + 2 = 6$
2. Rewrite: $24 \div 6 - 3 + 1$
3. Division: $24 \div 6 = 4$
4. Rewrite: $4 - 3 + 1$
5. Subtract: $4 - 3 = 1$
6. Add: $1 + 1 = 2$

Mathematical Concepts

PEMDAS/BODMAS: Parentheses/Brackets first, then Exponents/Orders, then Multiplication and Division (left to right), then Addition and Subtraction (left to right).

Common Mistakes

- Ignoring parentheses and dividing 24 by 4 first
- Performing operations in incorrect sequence
- Subtracting before adding

How to Avoid

Always start with parentheses. Work left to right for operations at the same level.

Alternative Method/Verification

Use a calculator: $24 \div (4 + 2) - 3 + 1 = 2$

Real-World Context

Budget calculation: Sharing \$24 among 6 people ($\$4 + 2$ people), then subtracting a \$3 fee and adding \$1 bonus gives \$2 per person.

Question 12: Converting Percentages

Correct Answer: \$37.50

Problem

A shirt costs \$50 with a 25% discount. What is the final price?

Step-by-Step Solution

1. Calculate the discount amount: $25\% \text{ of } \$50 = 0.25 \times 50 = 12.50$
2. Subtract from original price: $50 - 12.50 = 37.50$
3. Final price: \$37.50

Mathematical Concepts

A percentage discount reduces the original price. Formula: $\text{Final Price} = \text{Original Price} - (\text{Percentage} \times \text{Original Price})$

Common Mistakes

- Calculating 25% of 25 instead of 25% of 50
- Forgetting to subtract the discount from the original
- Using 250% instead of 0.25

How to Avoid

Convert percentage to decimal (divide by 100), multiply by the original amount, then subtract.

Alternative Method/Verification

If 25% off, you pay 75%: $0.75 \times 50 = 37.50$

Real-World Context

Shopping: A \$50 shirt with a 25% sale saves you \$12.50.

Question 13: Ratio and Proportion

Correct Answer: 12 cups flour

Problem

A recipe calls for flour to sugar in the ratio 3 : 2. If you use 8 cups of sugar, how much flour do you need?

Step-by-Step Solution

1. Set up the proportion: $\frac{3 \text{ flour}}{2 \text{ sugar}} = \frac{x \text{ flour}}{8 \text{ sugar}}$
2. Cross-multiply: $3 \times 8 = 2 \times x$
3. Simplify: $24 = 2x$
4. Solve for x : $x = \frac{24}{2} = 12$
5. Answer: 12 cups flour

Mathematical Concepts

Ratios compare quantities. A proportion states that two ratios are equal. Cross-multiplication solves proportions.

Common Mistakes

- Inverting the ratio parts
- Not cross-multiplying correctly
- Confusing the numerator and denominator placement

How to Avoid

Set up the proportion clearly with consistent placement. Cross-multiply: $\frac{a}{b} = \frac{c}{d}$ gives $ad = bc$.

Alternative Method/Verification

Scale factor: $8 \div 2 = 4$, so flour is $3 \times 4 = 12$

Real-World Context

Cooking: Scaling recipes using ratios ensures ingredients maintain proper proportions.

Question 14: Multi-Step Percentage

Correct Answer: \$216,000

Problem

A house costs \$200,000. After a 20% increase, then a 10% decrease, what is the final price?

Step-by-Step Solution

1. Calculate 20% increase: $20\% \text{ of } 200,000 = 0.20 \times 200,000 = 40,000$
2. New price after increase: $200,000 + 40,000 = 240,000$
3. Calculate 10% decrease: $10\% \text{ of } 240,000 = 0.10 \times 240,000 = 24,000$
4. Final price: $240,000 - 24,000 = 216,000$
5. Answer: \$216,000

Mathematical Concepts

When applying multiple percentage changes, apply them sequentially. The second percentage applies to the new amount, not the original.

Common Mistakes

- Adding 20% and subtracting 10% from the original (ignoring compounding)
- Calculating the 10% decrease from the original price
- Arithmetic errors in intermediate steps

How to Avoid

Work step by step. Always use the current amount as the base for the next percentage calculation.

Alternative Method/Verification

Multiply by factors: $200,000 \times 1.20 \times 0.90 = 216,000$

Real-World Context

Real estate: A house appreciates 20% then depreciates 10%, not returning to the original value.

Question 15: Fraction to Percent

Correct Answer: 60%

Problem

Convert $\frac{3}{5}$ to a percent.

Step-by-Step Solution

1. Use the formula: $\text{Percentage} = \frac{\text{Fraction}}{1} \times 100\%$
2. Substitute: $\frac{3}{5} \times 100\%$
3. Simplify: $\frac{3 \times 100}{5}\% = \frac{300}{5}\% = 60\%$

Mathematical Concepts

A fraction can be converted to a percentage by multiplying by 100. Alternatively, find an equivalent fraction with denominator 100.

Common Mistakes

- Forgetting to multiply by 100
- Dividing 100 by 5 and 3 separately
- Confusing the operation direction

How to Avoid

Always multiply the fraction by 100 to convert to percent.

Alternative Method/Verification

Equivalent fraction: $\frac{3}{5} = \frac{60}{100} = 60\%$

Real-World Context

Grades: Scoring $\frac{3}{5}$ questions correctly equals 60%.

Question 16: Negative Numbers

Correct Answer: -8

Problem

Calculate: $-3 - 5 = ?$

Step-by-Step Solution

1. Interpret subtraction as adding a negative: $-3 - 5 = -3 + (-5)$
2. Both numbers are negative; add their absolute values: $3 + 5 = 8$
3. Result is negative: -8
4. Answer: -8

Mathematical Concepts

Subtracting a positive number is the same as adding a negative. When adding two negatives, the result is negative with a magnitude equal to the sum of the absolute values.

Common Mistakes

- Getting -2 by subtracting 5 from 3 (ignoring the initial negative)
- Incorrect sign handling in intermediate steps

How to Avoid

Think of the number line: starting at -3 , move 5 units left to reach -8 .

Alternative Method/Verification

Use a number line or calculator: $-3 - 5 = -8$

Real-World Context

Temperature: If it's -3 and drops 5 more degrees, it becomes -8 .

Question 17: Multiplying Negatives

Correct Answer: -12

Problem

Calculate: $3 \times (-4) = ?$

Step-by-Step Solution

1. Recall the rule: positive times negative equals negative
2. Multiply absolute values: $3 \times 4 = 12$
3. Apply the sign: negative
4. Answer: -12

Mathematical Concepts

Sign rules for multiplication:

- Positive $\times$ Positive = Positive
- Positive $\times$ Negative = Negative
- Negative $\times$ Negative = Positive

Common Mistakes

- Treating the negative as a positive: $3 \times 4 = 12$
- Using an incorrect sign rule

How to Avoid

Remember: unlike signs give a negative product. Same signs give a positive product.

Alternative Method/Verification

Repeated addition: $3 \times (-4) = (-4) + (-4) + (-4) = -12$

Real-World Context

Debt: Having 3 debts of \$4 each equals a total debt of \$12 (negative balance).

Question 18: Multiple Fractions

Correct Answer: $\frac{2}{3}$

Problem

Calculate: $\frac{1}{2} + \frac{1}{6} = ?$

Step-by-Step Solution

1. Find LCD of 2 and 6: LCD = 6
2. Convert $\frac{1}{2}$ to sixths: $\frac{1}{2} = \frac{3}{6}$
3. Keep $\frac{1}{6}$ as is
4. Add: $\frac{3}{6} + \frac{1}{6} = \frac{4}{6}$
5. Simplify: $\frac{4}{6} = \frac{2}{3}$

Mathematical Concepts

LCD is the smallest number divisible by all denominators. Simplification reduces fractions to lowest terms by dividing by the GCD.

Common Mistakes

- Using 12 as the LCD instead of 6
- Forgetting to simplify the final answer
- Adding denominators

How to Avoid

Find the LCD by listing multiples. Always simplify the final fraction.

Alternative Method/Verification

Decimals: $0.5 + 0.167 = 0.667$, and $\frac{2}{3} = 0.667$

Real-World Context

Portions: $\frac{1}{2}$ pizza plus $\frac{1}{6}$ pizza equals $\frac{2}{3}$ pizza.

Question 19: Tax Calculation

Correct Answer: \$65.52

Problem

A book costs \$58.50. With a 12% sales tax, what is the total cost?

Step-by-Step Solution

1. Calculate tax amount: $12\% \text{ of } 58.50 = 0.12 \times 58.50 = 7.02$
2. Add tax to original price: $58.50 + 7.02 = 65.52$
3. Total: \$65.52

Mathematical Concepts

Sales tax is a percentage added to the original price. Formula: $\text{Total} = \text{Original Price} + (\text{Tax Rate} \times \text{Original Price})$

Common Mistakes

- Calculating 12% of 12 instead of 12% of 58.50
- Forgetting to add the tax to the original price
- Rounding prematurely

How to Avoid

Convert the percentage to decimal, multiply by the original amount, then add to find the total.

Alternative Method/Verification

Total is 112% of original: $1.12 \times 58.50 = 65.52$

Real-World Context

Shopping: Most purchases include sales tax, increasing the final price.

Question 20: Comparing Forms

Correct Answer: $\frac{3}{4} = 0.75 = 75\%$

Problem

Show that $\frac{3}{4}$, 0.75, and 75% are equivalent.

Step-by-Step Solution

1. Convert fraction to decimal: $\frac{3}{4} = 3 \div 4 = 0.75$
2. Convert decimal to percent: $0.75 \times 100\% = 75\%$
3. All three forms represent the same value

Mathematical Concepts

Fractions, decimals, and percentages are interchangeable representations of the same quantity.

Common Mistakes

- Converting fractions incorrectly to decimals
- Confusing decimal multiplication when converting to percent
- Not recognizing equivalent forms

How to Avoid

Practice converting between all three forms systematically.

Comparison Table

Fraction	Decimal	Percent
$\frac{1}{4}$	0.25	25%
$\frac{1}{2}$	0.50	50%
$\frac{3}{4}$	0.75	75%
$\frac{1}{5}$	0.20	20%
$\frac{2}{5}$	0.40	40%
$\frac{3}{5}$	0.60	60%

Real-World Context

Understanding different representations helps in interpreting data presented in various formats.

3 Advanced Level (Questions 21-30)

Question 21: Complex Fractions

Correct Answer: $\frac{19}{24}$

Problem

Calculate: $\frac{5}{8} + \frac{1}{6} = ?$

Step-by-Step Solution

1. Find LCD of 8 and 6:

Multiples of 8 : 8, 16, 24, 32, ...

Multiples of 6 : 6, 12, 18, 24, ...

$$\text{LCD} = 24$$

2. Convert $\frac{5}{8}$ to twenty-fourths: $\frac{5}{8} = \frac{5 \times 3}{8 \times 3} = \frac{15}{24}$

3. Convert $\frac{1}{6}$ to twenty-fourths: $\frac{1}{6} = \frac{1 \times 4}{6 \times 4} = \frac{4}{24}$

4. Add: $\frac{15}{24} + \frac{4}{24} = \frac{19}{24}$

5. Check if simplifiable: $\text{GCD}(19, 24) = 1$, so $\frac{19}{24}$ is in simplest form

Mathematical Concepts

Finding the LCD requires identifying the least common multiple of the denominators. Prime factorization can help find the LCD systematically.

Common Mistakes

- Using 48 as the LCD instead of 24
- Incorrect conversion of fractions
- Arithmetic errors in finding the LCD

How to Avoid

List multiples of each denominator until you find a common one. Use prime factorization for complex cases.

Alternative Method/Verification

Decimals: $0.625 + 0.167 = 0.792$, and $\frac{19}{24} \approx 0.792$

Real-World Context

Measurement: Combining $\frac{5}{8}$ inch and $\frac{1}{6}$ inch yields $\frac{19}{24}$ inch.

Question 22: Exponents

Correct Answer: 41

Problem

Calculate: $5^2 + 4^2 = ?$

Step-by-Step Solution

1. Evaluate 5^2 : $5 \times 5 = 25$
2. Evaluate 4^2 : $4 \times 4 = 16$
3. Add the results: $25 + 16 = 41$

Mathematical Concepts

Exponentiation is evaluated before addition. This relates to the Pythagorean theorem: $a^2 + b^2 = c^2$.

Common Mistakes

- Computing $5^2 + 4^2 = (5 + 4)^2 = 81$ (incorrect grouping)
- Multiplying instead of squaring
- Adding first, then squaring: $(5 + 4)^2 = 81$

How to Avoid

Evaluate exponents individually before performing other operations.

Alternative Method/Verification

Calculator: $5^2 + 4^2 = 25 + 16 = 41$

Real-World Context

Pythagorean theorem: A right triangle with legs of 5 and 4 units has a hypotenuse of $\sqrt{41}$ units.

Question 23: Square Roots

Correct Answer: 13 ft

Problem

A room has an area of 169 square feet. If the room is square-shaped, what is the length of one side?

Step-by-Step Solution

1. For a square, $\text{Area} = \text{side}^2$
2. Set up the equation: $\text{side}^2 = 169$
3. Take the square root: $\text{side} = \sqrt{169}$
4. Calculate: $\sqrt{169} = 13$ (since $13 \times 13 = 169$)
5. Answer: 13 feet

Mathematical Concepts

The square root is the inverse of squaring. For a square, the side length is the square root of the area.

Common Mistakes

- Dividing the area by 2: $169 \div 2 = 84.5$ (incorrect)
- Using $\frac{169}{4}$ as the side length
- Not recognizing 13 as the square root of 169

How to Avoid

Memorize perfect squares up to $20^2 = 400$.

Alternative Method/Verification

Check: $13 \times 13 = 169$

Real-World Context

Flooring: Determining room dimensions from area measurements requires taking square roots.

Question 24: Compound Percent

Correct Answer: \$115.56

Problem

An investment of \$100 earns 8% interest, then the new amount earns 7% interest. What is the final amount?

Step-by-Step Solution

1. Initial investment: \$100
2. After first year (8% interest): $100 \times 1.08 = 108$
3. After second year (7% on new amount): $108 \times 1.07 = 115.56$
4. Final amount: \$115.56

Mathematical Concepts

Compound interest applies interest to the accumulated amount, not the original. Formula: Final Amount = $P(1 + r_1)(1 + r_2)$

Common Mistakes

- Adding interest rates: $8\% + 7\% = 15\%$, then applying 15% to original
- Calculating the second interest on the original amount
- Rounding intermediate steps

How to Avoid

Apply each interest rate sequentially to the current amount.

Alternative Method/Verification

$$100 \times 1.08 \times 1.07 = 100 \times 1.1556 = 115.56$$

Real-World Context

Banking: Compound interest is how savings accounts and investments grow over multiple periods.

This completes the GED Math Answer Guide for Number Sense and Operations.